Garv Vaidya

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Education

Hitkarini College of Engineering and Technology — Jabalpur, MP
B.Tech in Computer Science & Engineering (RGPV, Bhopal) | CGPA: 7.17 / 10 — *2023 – 2027*

Technical Skills

AI/ML & Numerical Computing: PyTorch, Sentence-Transformers, NumPy (BLAS GEMV), HyDE, Google Cloud Vertex AI ADC, FinOps Metering, LLM APIs (Gemini, Groq)
Backend Systems & Concurrency: Python, Go, FastAPI, AsyncIO, Pydantic (V2), Flask, Uvicorn, Gunicorn, REST APIs, Microservices
Cloud Infrastructure & DevOps: AWS (EC2, SQS, RDS, S3, IAM, VPC), Docker, Kubernetes, Google gVisor (`runsc`), Terraform, LocalStack, GitHub Actions
Persistence, Security & Testing: PostgreSQL, SQL, Scrypt Hashing, Stateless JWT, Pytest, Git, Bash, Linux / POSIX

Projects

Ares – *Disaggregated Cloud-Native Vector Search Engine* — Deployed on Kubernetes | GitHub
Python, FastAPI, PyTorch, Sentence-Transformers, Docker, Kubernetes

- Architected a disaggregated vector search engine decoupling an asynchronous FastAPI coordinator from stateless compute shards via Kubernetes headless services, non-blocking CoreDNS dynamic TTL caching (5.0s in `run_in_executor`), and HPA targeting 50% CPU.
- Engineered vectorized cosine-similarity search over 384-dimensional dense embeddings (`all-MiniLM-L6-v2`), enforcing unit L_2 norm invariants at ingest to reduce retrieval to hardware-accelerated BLAS GEMV matrix dot products ($\mathbf{s} = M\mathbf{q}$).
- Integrated coordinator-isolated HyDE query expansion via Google Cloud Vertex AI ADC (`gemini-3.5-flash-lite`) and atomic index snapshotting (`.npz` via POSIX rename) to node-local mounts (`emptyDir`), achieving a verified p_{95} query latency of 596.81 ms under concurrent load (50 requests, 5 clients).

Leviathan – *Distributed DAG Workflow Orchestrator* — GitHub
Python, AsyncIO, AWS SQS, LocalStack, Terraform, Docker, Pydantic (V2)

- Engineered an asynchronous DAG task engine in Python 3.12 and AsyncIO, implementing Kahn's algorithm ($O(V+E)$ with `collections.deque`) for topological sorting and cycle detection alongside pre-flight Pydantic schema attribute validation.
- Decoupled task orchestration from worker execution using LocalStack AWS SQS and Terraform IaC, enforcing strict message response semantics with transient error retention and duplicate-delivery idempotency.
- Enforced Contract-First Pydantic V2 structured outputs via Vertex AI with automated validation retries, integrating exponential backoff with additive uniform jitter and cumulative FinOps token accounting.

SyncNote – *Multi-Tenant Cloud Note Service* — Deployed on AWS | GitHub
Python, Flask, PostgreSQL, Docker, AWS (EC2, RDS)

- Deployed a containerized RESTful service on AWS EC2 served by Gunicorn and connected to a managed AWS RDS PostgreSQL 16 instance over VPC security groups, implementing dual-mode 12-factor DSN configuration.
- Engineered database persistence utilizing process-level bounded connection pooling via `psycopg2` (1–10 sockets), eliminating per-request TCP handshakes and TLS overhead to achieve sub-50ms query responses.
- Enforced user tenant isolation across REST endpoints using Scrypt password hashing ($N = 32768$), stateless JWT bearer tokens, compound SQL IDOR prevention (`WHERE id = %s AND user_id = %s`), and zero-secret container rootfs isolation.

Lucid-CI – *AI-Native DevSecOps Platform & Dynamic Sandbox* — Deployed on AWS | GitHub
Go, Python, FastAPI, Docker, Google gVisor, Terraform, AWS (EC2, SQS), Vertex AI

- Architected an ephemeral dynamic execution sandbox on AWS EC2 using Docker and Google gVisor (`runsc`), enforcing syscall interception, strict cgroup quotas (512MB RAM, 100 PIDs), and zero-network isolation (`--net none`) to prevent container escapes.
- Built an automated remediation microservice using FastAPI and Vertex AI (`gemini-3.5-flash-lite`), implementing 128-bit cryptographic XML nonce fencing to neutralize prompt injection across 20 adversarial attack vectors (0.0% ASR).
- Engineered asynchronous PR scanning integrating a Go Tree-sitter taint engine with Pydantic V2 patch validation, automated failover to Gemini Flash, and modular Terraform provisioning for AWS SQS and VPC networks.

Certifications

Train A Small Language Model | Google DeepMind — Jun 2026
Deploy Multi-Agent Architectures | Google Cloud — Jun 2026
AWS Artificial Intelligence Practitioner | Amazon Web Services — May 2026